

# PROJECT DEVELOPMENT PHASE

## Machine Learning Model Performance Test

|               |                                                    |
|---------------|----------------------------------------------------|
| Date          | 02 October 2026                                    |
| Project Name  | Automated Network Request Management in ServiceNow |
| Maximum Marks | 10 Marks                                           |

### Model Performance Testing

Machine-learning performance metrics are not applicable to the current Automated Network Request Management project because it does not contain a regression or classification model. The current implementation is deterministic and configuration-driven in ServiceNow. Therefore, ML metrics are recorded as N/A rather than invented.

| S.No. | Parameter      | Values                                                                                                                                                                                                                                                                                              | Screenshot / Evidence                    |
|-------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1     | Metrics        | Regression Model: MAE - N/A, MSE - N/A, RMSE - N/A, R <sup>2</sup> - N/A.<br>Classification Model: Confusion Matrix - N/A, Accuracy Score - N/A,<br>Classification Report - N/A.<br>Reason: no trained ML model exists in the current project.                                                      | Not Applicable                           |
| 2     | Tune the Model | Hyperparameter Tuning - N/A<br>Validation Method - N/A<br>The project instead validates deterministic request-form behaviour, Flow Designer conditions, catalog-variable mappings, approval branching, notifications, request lifecycle processing, ACL/security behaviour, and workflow execution. | Use functional / UAT / workflow evidence |

### Alternative Performance Validation Used in This Project

- Request-form validation: verify the configured Network Request catalog variables and dynamic field behaviour.
- Workflow validation: verify Flow Designer retrieves submitted catalog variables and executes configured actions in sequence.
- Data validation: verify mapped values are stored correctly in Database Tables (u\_database\_tables).
- Approval validation: verify Approved and rejected outcomes follow the configured workflow branches.
- Notification validation: verify request-created and request-resolved notifications occur at configured points.
- Lifecycle validation: verify REQ/RITM records and request stages reflect the workflow outcome.
- Security validation: verify configured Create, Read, Write, and Delete ACL behaviour.
- Execution validation: use Flow Execution Details and platform logs to troubleshoot workflow actions, conditions, mappings, and notifications.

### Conclusion

No regression or classification model is implemented in the current ServiceNow solution, so MAE, MSE, RMSE, R<sup>2</sup>, confusion matrix, accuracy score, classification report, hyperparameter tuning, and ML validation metrics are N/A. Project performance is instead demonstrated through functional, workflow, data-mapping, approval, notification, security, lifecycle, and UAT evidence.